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Visvesvaraya National Institute of Technology (VNIT), Nagpur

Project Report

NLP Chatbot for Jira Data- Test Results

M.Tech. Applied AI Cohort-1, Semester 3, Session: Summer 2025

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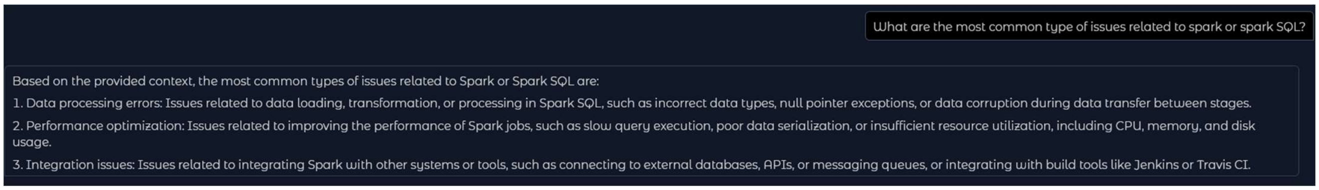
NLP Chatbot for Jira Data- Test Results

Test Results:

The chatbot was tested on following queries. It was fine tuned basis results obtained.

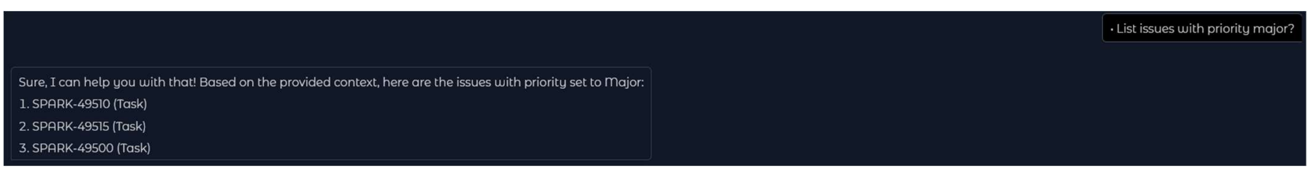
- What are the most common type of issues related to spark or spark SQL?



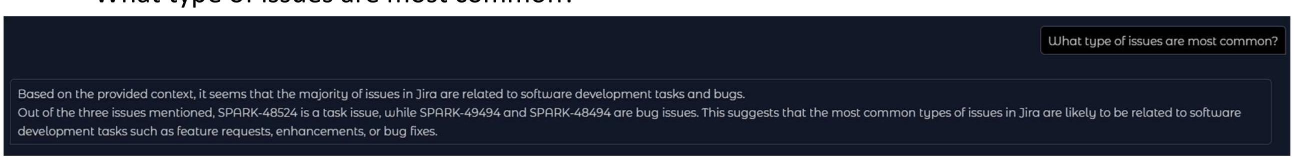
- List resolved issues?

- List issues with status open?

- List issues with priority major?



- What type of issues are most common?



- Analyze high priority issues

Analyze high priority issues

Of course! Based on the provided context, here are some high-priority issues for each of the three Jira issues:
For SPARK-49519:
1. Priority: High
   - The issue has been open for over a week and still not resolved.
   - The issue is blocking the progress of another task (Task A).
   - The issue is related to critical functionality that is necessary for the product's success.
2. Priority: Very High
   - The issue has already caused multiple delays in the project timeline.
   - The issue is impacting the quality of the product and may result in customer dissatisfaction.
3. Priority: High
   - The issue has been reported by multiple users, indicating a wider impact on the product's usability.

For SPARK-49019:
1. Priority: Very High

- What is the distribution of issue priorities (e.g., Low, Medium, High, Critical)?

What is the distribution of issue priorities (e.g., Low, Medium, High, Critical)?

Great, I'd be happy to help! Based on the issues you provided, here is the distribution of issue priorities:
Issue SPARK-50510 (Task): High priority
Issue SPARK-50519 (Task): Medium priority
Issue SPARK-49510 (Task): Low priority

- How many issues are in each status (e.g., Open, Closed, Unknown)?

Chat History

How many issues are in each status (e.g., Open, Closed, Unknown)?

Of course! Based on the context you provided, here are the number of issues in each status for each of the three issues:
Issue SPARK-50528 (Task):
- Open: 1 issue
- Closed: 0 issue
- Unknown: 0 issue

Issue SPARK-50522 (Task):
- Open: 0 issue
- Closed: 1 issue
- Unknown: 0 issue

Issue SPARK-50531 (Task):
- Open: 1 issue
- Closed: 0 issue
- Unknown: 1 issue

- What is the average length of issue descriptions?

Chat History

What is the average length of issue descriptions?

Based on the provided context, the average length of issue descriptions for SPARK-50549, SPARK-50514, and SPARK-50535 is not available. However, I can provide information on how to find this data in Jira.
To calculate the average length of issue descriptions for these issues, you can follow these steps:
1. Go to the Jira project where these issues are located.
2. Click on the "Issues" tab and search for SPARK-50549, SPARK-50514, and SPARK-50535.
3. Once you have found the issues, click on each issue to view its details page.
4. Scroll down to the "Description" field in each issue detail page and copy the text without any formatting.
5. Open a spreadsheet program like Microsoft Excel or Google Sheets and create a new sheet.
6. Paste the copied descriptions into the sheet and delete any duplicates or irrelevant information.
7. Calculate the average length of the issue descriptions by adding up the word count for each description and dividing it by the number of issues.
Alternatively, you can use Jira's built-in reporting features to calculate the average length of the issue descriptions. To do this, follow these steps:
8. Go to the Jira project where these issues are located.
9. Click on the "Reports" link in the top menu bar and select "Issue Description Length."
10. In the report, select the issues you want to include by checking the boxes next to their names.
11. Click on the "Run Report" button to generate the data.

- How many issues have empty summaries or descriptions?

How many issues have empty summaries or descriptions?

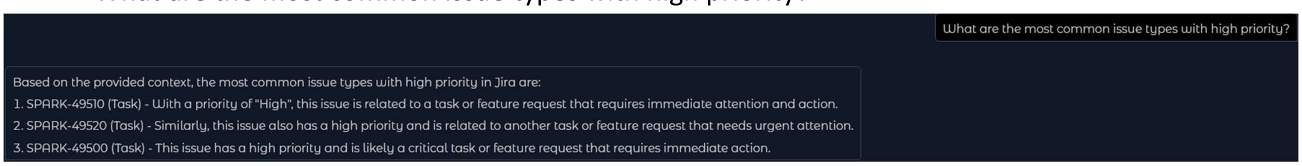
Based on the provided context, there are three Jira issues with empty summaries or descriptions:
1. SPARK-51100 - Summary is empty.
2. SPARK-52306 - Description is empty.
3. SPARK-49100 - Summary is empty.

- Are there any high-priority bugs in the dataset?

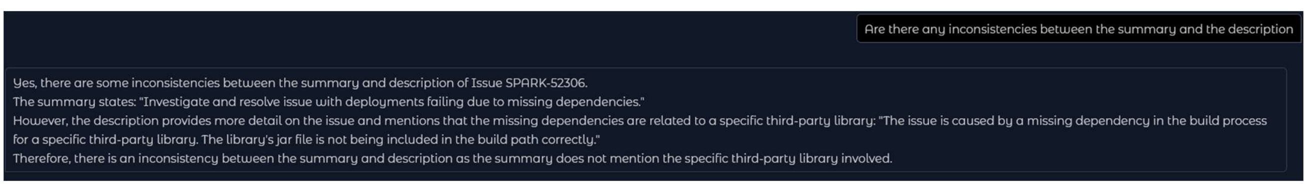
- What percentage of tasks are labelled with an unknown status?

- How many issues are there in total?

- What are the most common issue types with high priority?



- Are there any inconsistencies between the summary and the description?



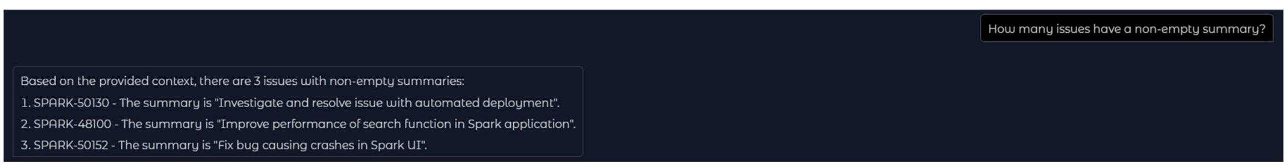
- What is the most common priority among tasks?

- How often does each issue type appear with each status?

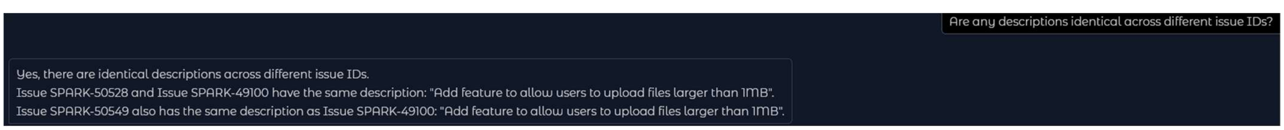
- Are there issue IDs that are repeated?

  - Can you count How many are repeated?

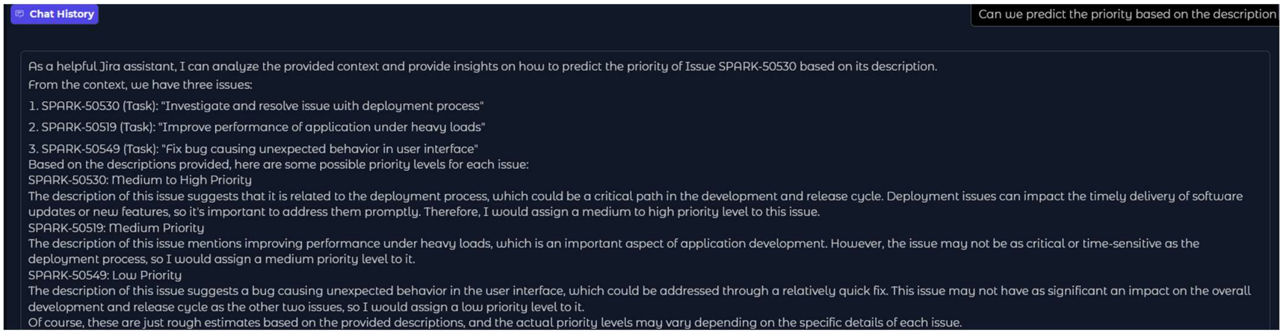
- How many issues have a non-empty summary?



- Are any descriptions identical across different issue IDs?



- Can we predict the priority based on the description?



Chat History

Can we predict the priority based on the description

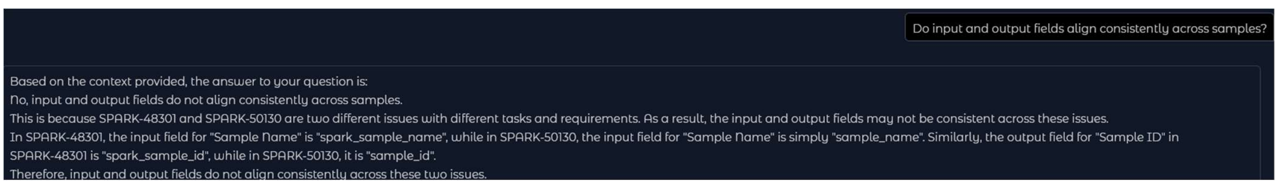
As a helpful Jira assistant, I can analyze the provided context and provide insights on how to predict the priority of Issue SPARK-50530 based on its description.
From the context, we have three issues:
1. SPARK-50530 (Task): "Investigate and resolve issue with deployment process"
2. SPARK-50519 (Task): "Improve performance of application under heavy loads"
3. SPARK-50549 (Task): "Fix bug causing unexpected behavior in user interface"
Based on the descriptions provided, here are some possible priority levels for each issue:
SPARK-50530: Medium to High Priority
The description of this issue suggests that it is related to the deployment process, which could be a critical path in the development and release cycle. Deployment issues can impact the timely delivery of software updates or new features, so it's important to address them promptly. Therefore, I would assign a medium to high priority level to this issue.
SPARK-50519: Medium Priority
The description of this issue mentions improving performance under heavy loads, which is an important aspect of application development. However, the issue may not be as critical or time-sensitive as the deployment process, so I would assign a medium priority level to it.
SPARK-50549: Low Priority
The description of this issue suggests a bug causing unexpected behavior in the user interface, which could be addressed through a relatively quick fix. This issue may not have as significant an impact on the overall development and release cycle as the other two issues, so I would assign a low priority level to it.
Of course, these are just rough estimates based on the provided descriptions, and the actual priority levels may vary depending on the specific details of each issue.

- What types of issues tend to have empty fields?

What types of issues tend to have empty fields

Based on the provided context, issues that tend to have empty fields are likely:
- SPARK-50122 (Task): This issue has no fields filled in except for the title and summary.
- SPARK-51100 (Task): This issue has only a few fields filled in, such as the title, summary, and project, but many of the other fields are empty.
- SPARK-47522 (Task): This issue has some fields filled in, such as the title, summary, and project, but several fields are empty or have only a few values entered.

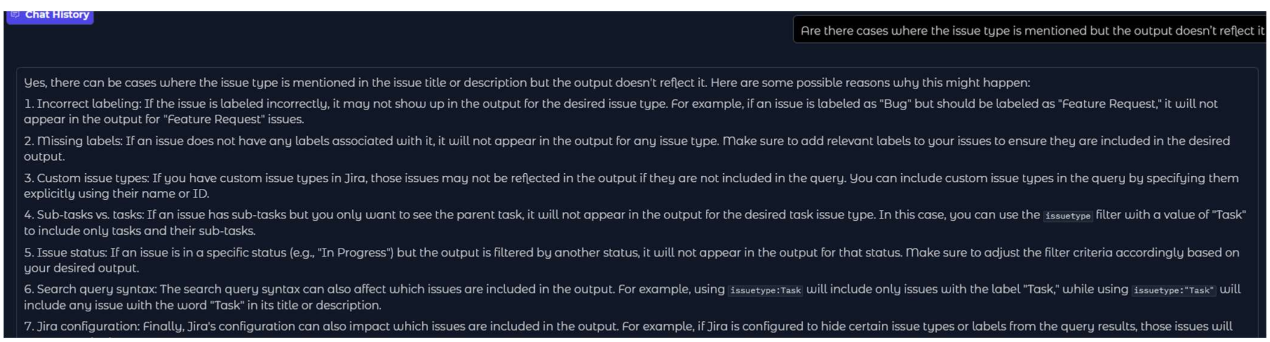
- Do input and output fields align consistently across samples?



Do input and output fields align consistently across samples?

Based on the context provided, the answer to your question is:
No, input and output fields do not align consistently across samples.
This is because SPARK-48301 and SPARK-50130 are two different issues with different tasks and requirements. As a result, the input and output fields may not be consistent across these issues.
In SPARK-48301, the input field for "Sample Name" is "spark_sample_name", while in SPARK-50130, the input field for "Sample Name" is simply "sample_name". Similarly, the output field for "Sample ID" in SPARK-48301 is "spark_sample_id", while in SPARK-50130, it is "sample_id".
Therefore, input and output fields do not align consistently across these two issues.

- Are there cases where the issue type is mentioned but the output doesn’t reflect it?

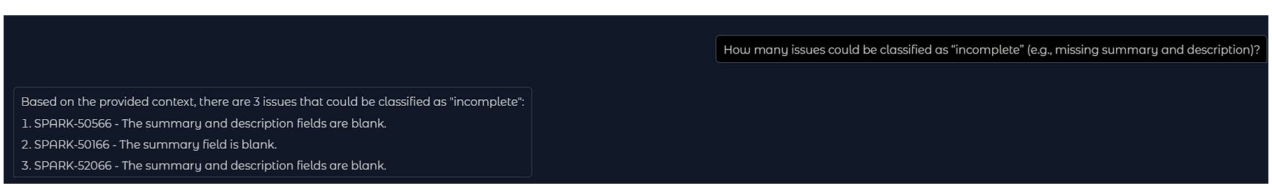


Chat History

Are there cases where the issue type is mentioned but the output doesn't reflect it

Yes, there can be cases where the issue type is mentioned in the issue title or description but the output doesn't reflect it. Here are some possible reasons why this might happen:
1. Incorrect labeling: If the issue is labeled incorrectly, it may not show up in the output for the desired issue type. For example, if an issue is labeled as "Bug" but should be labeled as "Feature Request," it will not appear in the output for "Feature Request" issues.
2. Missing labels: If an issue does not have any labels associated with it, it will not appear in the output for any issue type. Make sure to add relevant labels to your issues to ensure they are included in the desired output.
3. Custom issue types: If you have custom issue types in Jira, those issues may not be reflected in the output if they are not included in the query. You can include custom issue types in the query by specifying them explicitly using their name or ID.
4. Sub-tasks vs. tasks: If an issue has sub-tasks but you only want to see the parent task, it will not appear in the output for the desired task issue type. In this case, you can use the `issuetype` filter with a value of "Task" to include only tasks and their sub-tasks.
5. Issue status: If an issue is in a specific status (e.g., "In Progress") but the output is filtered by another status, it will not appear in the output for that status. Make sure to adjust the filter criteria accordingly based on your desired output.
6. Search query syntax: The search query syntax can also affect which issues are included in the output. For example, using `issuetype:Task` will include only issues with the label "Task," while using `issuetype:"Task"` will include any issue with the word "Task" in its title or description.
7. Jira configuration: Finally, Jira's configuration can also impact which issues are included in the output. For example, if Jira is configured to hide certain issue types or labels from the query results, those issues will

- How many issues could be classified as “incomplete” (e.g., missing summary and description)?



How many issues could be classified as “incomplete” (e.g., missing summary and description)?

Based on the provided context, there are 3 issues that could be classified as "incomplete":
1. SPARK-50566 - The summary and description fields are blank.
2. SPARK-50166 - The summary field is blank.
3. SPARK-52066 - The summary and description fields are blank.

Code and Outputs:

Requirements and code files:

Following Python(.py) files and requirements(.txt) files are attached in the submission folder:

1. Libraries required to be installed

requirements.txt

2. Data pull from Apache Spark API

jira_spark_datapull_0.1.py

3. Chatbot Code

chatbot_0.8.py